Affordable Non-Contact Respiration Monitoring Using Wi-Fi CSI and Doppler Radar: A Performance-Cost Trade-Off for Resource-Limited Settings

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2025 Asia-Pacific Microwave Conference (APMC) | 979-8-3315-3455-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/APMC65046.2025.11377708

***Abstract*— Non-contact vital sign monitoring has gained significant attention for enabling continuous in-home health surveillance without user discomfort or privacy intrusion. This study compares the performance and economic feasibility of two promising technologies—Wi-Fi Channel State Information (CSI) and Doppler radar phase measurement—for non-contact respiration monitoring. We developed and evaluated a system integrating a commodity 2.4 GHz Wi-Fi module and a 24 GHz continuous-wave (CW) Doppler radar, collecting respiration data from ten participants. The systems achieved a mean breathing rate accuracy of 92.01% (Wi-Fi) and 93.95% (radar), validated against a medical-grade chest belt. A cost-performance analysis indicates that Wi-Fi sensing offers a highly cost-effective solution, providing comparable accuracy to radar at a fraction of the hardware cost, making it suitable for scalable deployment in low-income regions, such as Bangladesh.**

I. Introduction

Respiratory monitoring is critical for detecting and managing conditions such as sleep apnea, asthma, and chronic obstructive pulmonary disease (COPD) [1-2]. In low and middle-income countries (LMICs) like Bangladesh, the burden of respiratory diseases remains high, while cost and infrastructure constraints limit access to continuous monitoring technologies [2-5]. However, continuous breathing monitoring, which can be crucial for treating these diseases, is hindered by resource constraints in those countries. Therefore, inexpensive methods for constant monitoring in such contexts are essential. In recent years, remote breath rate monitoring technologies have seen significant advancements, particularly following the COVID-19 pandemic, which highlighted the importance of non-contact, continuous respiratory monitoring [6]. Vision-based sensing for vital signs monitoring using a camera is gaining attention, but it remains a concern due to privacy concerns [7]. Radio-frequency (RF) technologies, including Wi-Fi [8] and radar, are gaining more attention than vision-based sensing due to their unobtrusive and non-invasive measurement nature. Non-contact radio-frequency (RF) sensing methods, including Wi-Fi CSI and radar, offer unobtrusive and privacy-preserving solutions. Despite independent demonstrations of their efficacy [6-9], Comparative studies examining the technical performance and economic feasibility of these modalities are currently unexplored, particularly in resource-limited settings.

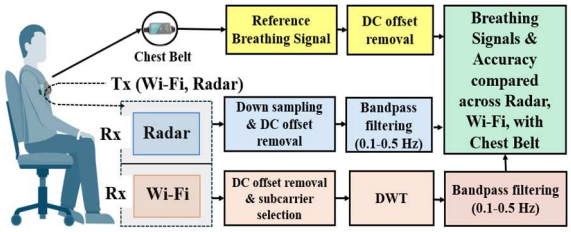

Fig. 1. Block diagram of the proposed respiration monitoring system using Wi-Fi CSI and radar. The subject's chest wall continuously reflects data packets and signals transmitted from the Tx. The received signals are processed to extract the respiration rate, which is then compared against a gold-standard reference.

In this work, we have performed system-level integration of the 2.4-GHz Wi-Fi module to extract channel state information (CSI) and the 24-GHz continuous wave (CW) radar module to extract vital signs from ten different participants concurrently. A signal-processing algorithm was also developed and integrated with the hardware system to monitor vital signs. After extracting vital signs, they were compared with the traditional ground truth reference sensor (Go Direct Respiration Belt, Vernier Science Education, USA). This study also focused on evaluating the accuracy and cost trade-offs between Wi-Fi CSI-based and radar-based respiration monitoring systems. We aim to identify solutions that can be scaled for health monitoring applications in Bangladesh and similar settings. **Fig. 1** illustrates the proposed block diagram of a vital signs monitoring system for a static subject, utilizing Wi-Fi CSI and CW radar.

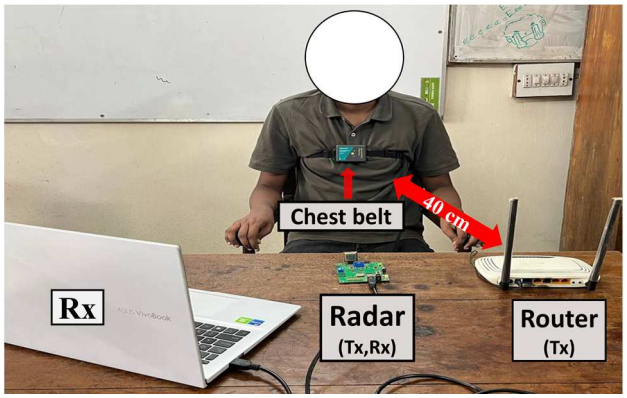

Fig. 2. Experimental setup for data collection using the Wi-Fi router, radar, and the reference chest belt, with the subject seated at 40 cm from the router.

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II. Theoretical Background

A. WI-FI CSI

In Wi-Fi sensing, the CSI captures motion-induced phase and amplitude variations, enabling the detection of such physiological activities by analyzing these variations [10]. Orthogonal Frequency Division Multiplexing (OFDM) is a modulation technique used in Wi-Fi standards like IEEE 802.11ac [10]. It divides the channel spectrum into subcarriers [11], transmitting data via Inverse Fast Fourier Transform (IFFT) to counter frequency-selective fading. CSI can be represented as:

$$Y(t) = H(f,t)X(f,t) + N(f,t) \quad (1)$$

Where X and Y are transmitted and received signal, N is noise, and H is the CSI matrix. H can be written as m×n×w, where m is the number of transmitter antennas, n is the number of receiver antennas, and w is the number of subcarriers. Each matrix element can be represented as:

$$h_k = |h_k| e^{j\sin\theta} \quad (2)$$

Where $|h_k|$ is the amplitude and θ is the phase of the relevant subcarriers.

B. Radar

According to Doppler theory, the phase information of reflected echoes changes when a transmitted signal encounters a moving target, such as a human. The phase change in the reflected signal is directly proportional to the tiny movement of the chest surface due to cardiac-respiratory activities, such as breathing [12]. The phase change of the reflected echo can be represented as:

$$\Theta \propto \frac{4\pi X(t)}{\lambda} \quad (3)$$

Where $\Theta(t)$ is the phase of the reflected signal, $X(t)$ is the periodic movement of the chest wall due to cardiorespiratory activities. The breathing rate can be extracted from the phase change information of the reflected signal.

III. Materials and Methods

A. Experimental Setup & Data Acquisition

A TP-Link Archer C6 router (IEEE 802.11ac, 2.4GHz) was used as the transmitter, with an ASUS X512FL laptop and Intel AX200 NIC as the receiver. A 24 GHz K-LC1a CW Doppler radar module from RFbeam Microwave, along with one ST100 starter kit was employed for the radar system. For Wi-Fi CSI extraction, PicoScenes was used as middleware for MATLAB platforms [13]. Reference data was obtained using the Vernier Go Direct Respiration Belt with Vernier Graphical Analysis software (Vernier, Beaverton, USA). **Fig.2** illustrates the experimental setup and the synchronized data acquisition process for all three modalities. The experiments involved 10 healthy subjects (aged 21–25) seated still in a controlled environment. Each wore a chest belt, and data was recorded for 1 minute per trial, totaling 10 minutes (~160 breathing cycles) for Wi-Fi, radar, and chest belt. Subjects were positioned 40 cm from the system, with the receiver laptop placed 10 cm behind the transmitters. Ethical approval was obtained from the Ethical Review Committee, Faculty of Biological Sciences, University of Dhaka (Application No. 18), in accordance with the guidelines of the AEERC.

According to the market analysis presented in **Table I**, based on vendor datasheets and distributors, radar systems are costly due to their specialized hardware although they offer high accuracy whereas Wi-Fi-based systems reduce system costs by approximately 96-98%, depending on the chosen Wi-Fi hardware. Moreover, Wi-Fi routers are already deployed in homes for communication, without requiring any additional hardware for sensing. This highlights a performance-cost trade off establishing Wi-Fi systems as a scalable solution for third-world countries.

TABLE I

COST COMPARISON BETWEEN BOTH SYSTEMS

PRICES ESTIMATED FROM VENDOR DATASHEETS AND DISTRIBUTOR LISTINGS FOR WI-FI AND RADAR SYSTEMS.

Wi-Fi Systems	Cost	Radar systems	Cost
Intel AX200 NIC+ software [15]	~$11	RFbeam K-LC1a + ST100 Starter Kit [16]	~$564
ESP32 + CSI Firmware [17]	~$20–78	TI IWR1443BOOST [18]	~$305

B. Signal Processing And Subcarrier Selection

The PicoScenes Wi-Fi AP mode was used to capture CSI data, and the collected raw data files were imported into MATLAB for analysis. The raw CSI files were parsed in MATLAB using the PicoScenes MATLAB Toolbox Core (PMT Core), and the CSI amplitude was extracted. Then, the DC offset was removed, and the SNR was computed for all 245 subcarriers. The subcarriers were sorted, and the top 30 subcarriers were selected to reduce processing complexity, as it was calculated that these subcarriers retained 78% of the total SNR of all subcarriers. Then, denoising was performed using a third-level discrete wavelet transform (Daubechies) [14].

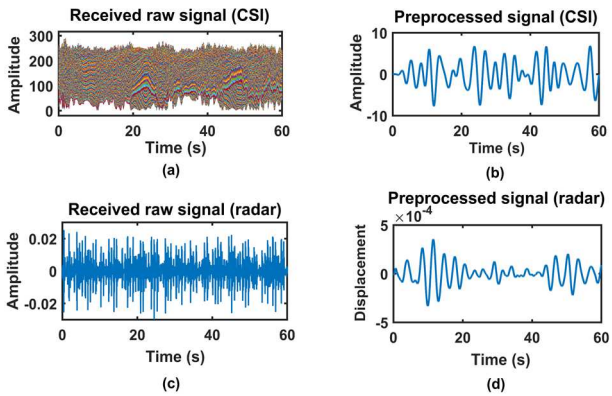

Fig. 3. Time domain representation of (a) raw CSI and (b) processed and denoised CSI with (c) raw radar waveform and (d) preprocessed radar waveform of the subject.

Following, a 3rd-order Butterworth bandpass filter (0.1–0.5 Hz) was employed to extract the respiration signal. Then, an FFT was performed to extract the respiration rate. For the radar signal, the raw data was first down-sampled to 100 Hz, and the DC offset was removed. Subsequently, a 3rd-order Butterworth bandpass filter with a cutoff frequency of 0.1–0.5 Hz was applied to extract the breathing waveform. After that, the filtered radar data's Fast Fourier Transform (FFT)

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was performed to extract the respiration rate. Similarly, the respiration belt signal was also processed. The raw and preprocessed signals are shown on **Fig. 3**.

IV. Results

The extracted breathing waveforms collected from Wi-Fi and radar are compared with reference data in both the time and

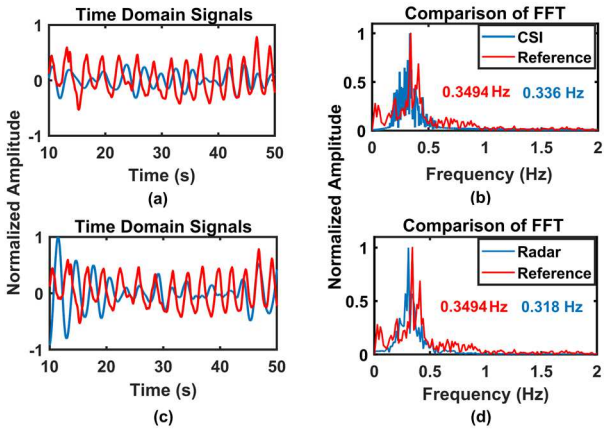

Fig. 4. Time domain representation of the respiration waveforms of a subject collected from (a)Wi-Fi and respiration belt and (c) Radar and Respiration Belt and their comparison of FFTs with (b)Wi-Fi and (d) radar respectively.

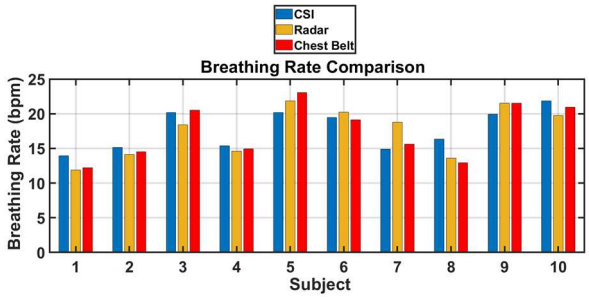

Fig. 5. Comparison of respiration rate estimation across 10 subjects using CSI, radar and chest belt

frequency domains. For subject 3, **Fig. 4** shows that the breathing frequencies extracted from the chest belt, Wi-Fi, and radar were 0.3494 Hz, 0.336 Hz (Fig. 4(b)), and 0.31 Hz (Fig. 4(d), respectively. **Fig. 5** depicts the respiration rates for all 10 subjects using the three modalities. The BR accuracy was then calculated using the following equation:

$$\text{Accuracy}(\%) = \left[1 - \frac{|\text{Reference BR(chestbelt)} - \text{Extracted BR}|}{\text{Reference BR(chestbelt)}}\right] \times 100 \quad (4)$$

The radar system achieved a mean accuracy of 93.95%, while Wi-Fi reached 92.01% across 10 subjects, indicating similar performance across both modalities.

V. Conclusion

This study evaluates non-contact respiration monitoring using Doppler radar and CSI from low-cost Wi-Fi devices, achieving accuracies of 92.01% and 93.95%, respectively, thereby validating both modalities for reliable detection of vital signs. Both Wi-Fi CSI and Doppler radar demonstrate accurate non-contact respiration monitoring, with Wi-Fi offering a cost-effective, scalable solution for resource-constrained environments, such as those found in Bangladesh. This work supports low-cost remote health monitoring with future works targeting population-scale deployment and expanded vital-sign detection capabilities.

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